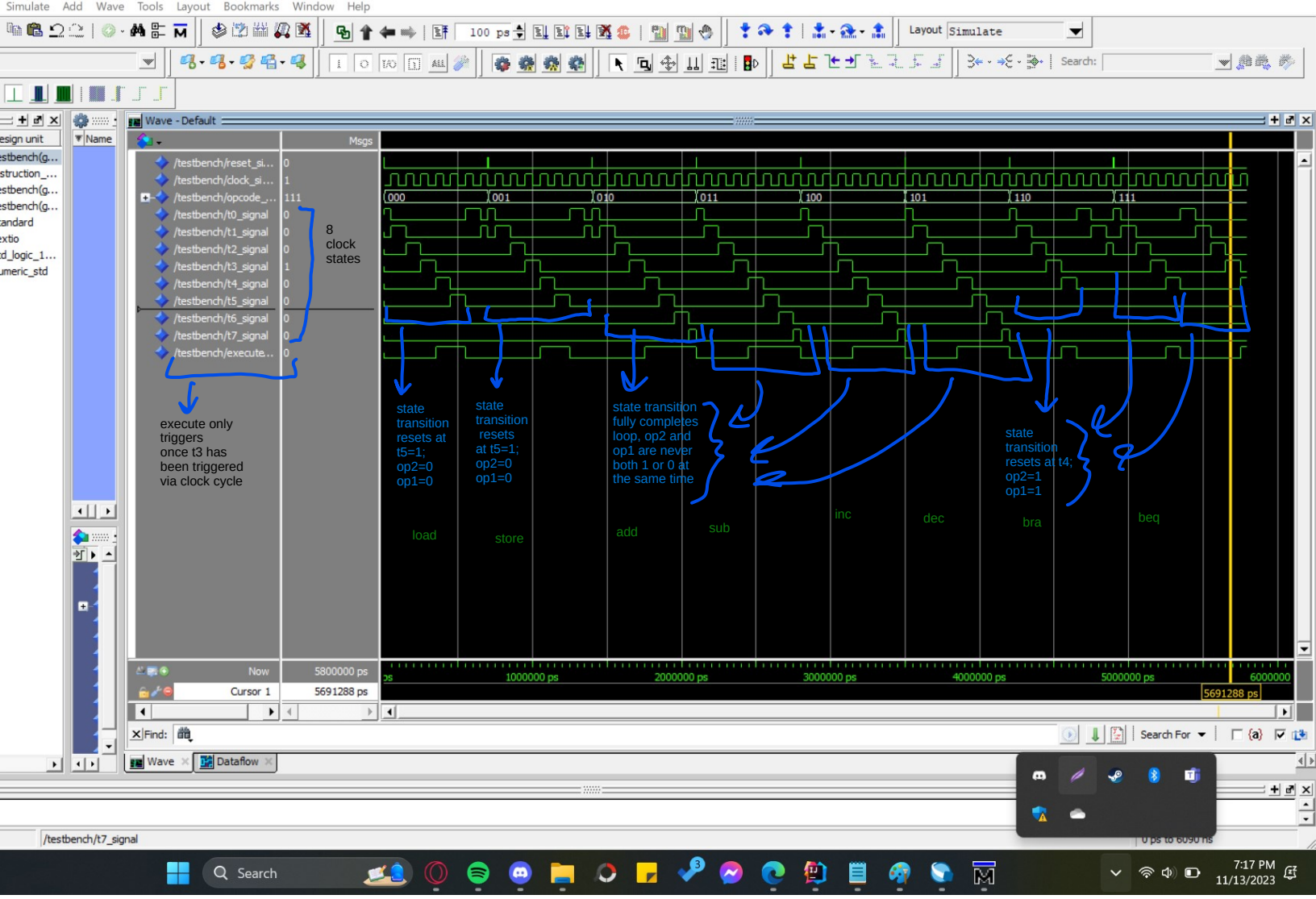
Wave - Default
/testbench/reset_si...
/testbench/clock_si...
/testbench/opcode_...
/testbench/t0_signal
/testbench/t1_signal
/testbench/t2_signal
/testbench/t3_signal
/testbench/t4_signal
/testbench/t5_signal
/testbench/t6_signal
/testbench/t7_signal
/testbench/execute...
000
001
010
011
100
101
110
111
8 clock states
execute only triggers once t3 has been triggered via clock cycle
state transition resets at t5=1; op2=0 op1=0
state transition resets at t5=1; op2=0 op1=0
state transition fully completes loop, op2 and op1 are never both 1 or 0 at the same time
state transition resets at t4; op2=1 op1=1
load
store
add
sub
inc
dec
bra
beq
Now
5800000 ps
Cursor 1
5691288 ps
1000000 ps
2000000 ps
3000000 ps
4000000 ps
5000000 ps
6000000
Wave
Dataflow
/testbench/t7_signal

```
# Loading work.d_ff(rtl)
# ** Note: T0 correct for current state = 000
#    Time: 3 ns  Iteration: 0  Instance: /testbench
# ** Note: Execute correct for current state = 000
#    Time: 3 ns  Iteration: 0  Instance: /testbench
# ** Note: T1 correct for current state = 001
#    Time: 100 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 001
#    Time: 100 ns  Iteration: 1  Instance: /testbench
# ** Note: T2 correct for current state = 010
#    Time: 200 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 010
#    Time: 200 ns  Iteration: 1  Instance: /testbench
# ** Note: T3 correct for current state = 011
#    Time: 300 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 011
#    Time: 300 ns  Iteration: 1  Instance: /testbench
# ** Note: T4 correct for current state = 100
#    Time: 400 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 100
#    Time: 400 ns  Iteration: 1  Instance: /testbench
# ** Note: T5 correct for current state = 101
#    Time: 500 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 101
#    Time: 500 ns  Iteration: 1  Instance: /testbench
# ** Note: T0 correct for current state = 000
#    Time: 600 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 000
#    Time: 600 ns  Iteration: 1  Instance: /testbench
# ** Note: T1 correct for current state = 001
#    Time: 700 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 001
#    Time: 700 ns  Iteration: 1  Instance: /testbench
# ** Note: T0 correct for current state = 000
#    Time: 703 ns  Iteration: 0  Instance: /testbench
# ** Note: Execute correct for current state = 000
#    Time: 703 ns  Iteration: 0  Instance: /testbench
# ** Note: T1 correct for current state = 001
#    Time: 800 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 001
#    Time: 800 ns  Iteration: 1  Instance: /testbench
# ** Note: T2 correct for current state = 010
#    Time: 900 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 010
```

```
# ** Note: T3 correct for current state = 011
#    Time: 1 us  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 011
#    Time: 1 us  Iteration: 1  Instance: /testbench
# ** Note: T4 correct for current state = 100
#    Time: 1100 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 100
#    Time: 1100 ns  Iteration: 1  Instance: /testbench
# ** Note: T5 correct for current state = 101
#    Time: 1200 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 101
#    Time: 1200 ns  Iteration: 1  Instance: /testbench
# ** Note: T0 correct for current state = 000
#    Time: 1300 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 000
#    Time: 1300 ns  Iteration: 1  Instance: /testbench
# ** Note: T1 correct for current state = 001
#    Time: 1400 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 001
#    Time: 1400 ns  Iteration: 1  Instance: /testbench
# ** Note: T0 correct for current state = 000
#    Time: 1403 ns  Iteration: 0  Instance: /testbench
# ** Note: Execute correct for current state = 000
#    Time: 1403 ns  Iteration: 0  Instance: /testbench
# ** Note: T1 correct for current state = 001
#    Time: 1500 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 001
#    Time: 1500 ns  Iteration: 1  Instance: /testbench
# ** Note: T2 correct for current state = 010
#    Time: 1600 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 010
#    Time: 1600 ns  Iteration: 1  Instance: /testbench
# ** Note: T3 correct for current state = 011
#    Time: 1700 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 011
#    Time: 1700 ns  Iteration: 1  Instance: /testbench
# ** Note: T4 correct for current state = 100
#    Time: 1800 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 100
#    Time: 1800 ns  Iteration: 1  Instance: /testbench
# ** Note: T5 correct for current state = 101
#    Time: 1900 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 101
#    Time: 1900 ns  Iteration: 1  Instance: /testbench
```

```
# ** Note: Execute correct for current state = 101
#    Time: 1900 ns  Iteration: 1  Instance: /testbench
# ** Note: T6 correct for current state = 110
#    Time: 2 us  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 110
#    Time: 2 us  Iteration: 1  Instance: /testbench
# ** Note: T7 correct for current state = 111
#    Time: 2100 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 111
#    Time: 2100 ns  Iteration: 1  Instance: /testbench
# ** Note: T0 correct for current state = 000
#    Time: 2103 ns  Iteration: 0  Instance: /testbench
# ** Note: Execute correct for current state = 000
#    Time: 2103 ns  Iteration: 0  Instance: /testbench
# ** Note: T1 correct for current state = 001
#    Time: 2200 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 001
#    Time: 2200 ns  Iteration: 1  Instance: /testbench
# ** Note: T2 correct for current state = 010
#    Time: 2300 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 010
#    Time: 2300 ns  Iteration: 1  Instance: /testbench
# ** Note: T3 correct for current state = 011
#    Time: 2400 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 011
#    Time: 2400 ns  Iteration: 1  Instance: /testbench
# ** Note: T4 correct for current state = 100
#    Time: 2500 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 100
#    Time: 2500 ns  Iteration: 1  Instance: /testbench
# ** Note: T5 correct for current state = 101
#    Time: 2600 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 101
#    Time: 2600 ns  Iteration: 1  Instance: /testbench
# ** Note: T6 correct for current state = 110
#    Time: 2700 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 110
#    Time: 2700 ns  Iteration: 1  Instance: /testbench
# ** Note: T7 correct for current state = 111
#    Time: 2800 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 111
#    Time: 2800 ns  Iteration: 1  Instance: /testbench
# ** Note: T0 correct for current state = 000
#    Time: 2803 ns  Iteration: 0  Instance: /testbench
# ** Note: Execute correct for current state = 000
#    Time: 2803 ns  Iteration: 0  Instance: /testbench
# ** Note: T1 correct for current state = 001
#    Time: 2900 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 001
#    Time: 2900 ns  Iteration: 1  Instance: /testbench
# ** Note: T2 correct for current state = 010
#    Time: 3 us  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 010
```

```
# ** Note: T5 correct for current state = 101
#    Time: 3300 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 101
#    Time: 3300 ns  Iteration: 1  Instance: /testbench
# ** Note: T6 correct for current state = 110
#    Time: 3400 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 110
#    Time: 3400 ns  Iteration: 1  Instance: /testbench
# ** Note: T7 correct for current state = 111
#    Time: 3500 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 111
#    Time: 3500 ns  Iteration: 1  Instance: /testbench
# ** Note: T0 correct for current state = 000
#    Time: 3503 ns  Iteration: 0  Instance: /testbench
# ** Note: Execute correct for current state = 000
#    Time: 3503 ns  Iteration: 0  Instance: /testbench
# ** Note: T1 correct for current state = 001
#    Time: 3600 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 001
#    Time: 3600 ns  Iteration: 1  Instance: /testbench
# ** Note: T2 correct for current state = 010
#    Time: 3700 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 010
#    Time: 3700 ns  Iteration: 1  Instance: /testbench
# ** Note: T3 correct for current state = 011
#    Time: 3800 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 011
#    Time: 3800 ns  Iteration: 1  Instance: /testbench
# ** Note: T4 correct for current state = 100
#    Time: 3900 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 100
#    Time: 3900 ns  Iteration: 1  Instance: /testbench
# ** Note: T5 correct for current state = 101
#    Time: 4 us  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 101
#    Time: 4 us  Iteration: 1  Instance: /testbench
# ** Note: T6 correct for current state = 110
#    Time: 4100 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 110
#    Time: 4100 ns  Iteration: 1  Instance: /testbench
# ** Note: T7 correct for current state = 111
#    Time: 4200 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 111
#    Time: 4200 ns  Iteration: 1  Instance: /testbench
# ** Note: T0 correct for current state = 000
#    Time: 4203 ns  Iteration: 0  Instance: /testbench
# ** Note: Execute correct for current state = 000
#    Time: 4203 ns  Iteration: 0  Instance: /testbench
# ** Note: T1 correct for current state = 001
#    Time: 4300 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 001
#    Time: 4300 ns  Iteration: 1  Instance: /testbench
# ** Note: T2 correct for current state = 010
```

```
#    Time: 4500 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 011
#    Time: 4500 ns  Iteration: 1  Instance: /testbench
# ** Note: T4 correct for current state = 100
#    Time: 4600 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 100
#    Time: 4600 ns  Iteration: 1  Instance: /testbench
# ** Note: T0 correct for current state = 000
#    Time: 4700 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 000
#    Time: 4700 ns  Iteration: 1  Instance: /testbench
# ** Note: T1 correct for current state = 001
#    Time: 4800 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 001
#    Time: 4800 ns  Iteration: 1  Instance: /testbench
# ** Note: T2 correct for current state = 010
#    Time: 4900 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 010
#    Time: 4900 ns  Iteration: 1  Instance: /testbench
# ** Note: T0 correct for current state = 000
#    Time: 4903 ns  Iteration: 0  Instance: /testbench
# ** Note: Execute correct for current state = 000
#    Time: 4903 ns  Iteration: 0  Instance: /testbench
# ** Note: T1 correct for current state = 001
#    Time: 5 us  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 001
#    Time: 5 us  Iteration: 1  Instance: /testbench
# ** Note: T2 correct for current state = 010
#    Time: 5100 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 010
#    Time: 5100 ns  Iteration: 1  Instance: /testbench
# ** Note: T3 correct for current state = 011
#    Time: 5200 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 011
#    Time: 5200 ns  Iteration: 1  Instance: /testbench
# ** Note: T4 correct for current state = 100
#    Time: 5300 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 100
#    Time: 5300 ns  Iteration: 1  Instance: /testbench
# ** Note: T0 correct for current state = 000
#    Time: 5400 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 000
#    Time: 5400 ns  Iteration: 1  Instance: /testbench
# ** Note: T1 correct for current state = 001
#    Time: 5500 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 001
#    Time: 5500 ns  Iteration: 1  Instance: /testbench
# ** Note: T2 correct for current state = 010
#    Time: 5600 ns  Iteration: 1  Instance: /testbench
# ** Note: Execute correct for current state = 010
#    Time: 5600 ns  Iteration: 1  Instance: /testbench
```